Presentation Outline: Does Air Quality Explain Adult Asthma?

SCQA

- **Situation:** Asthma affects millions of U.S. adults, and air pollution is a well-known trigger. The Los Angeles area has long faced air-quality challenges.
- **Complication:** If cleaner air means less asthma, air quality could guide where to focus prevention. But it is unclear whether county-level air quality actually tracks asthma rates.
- **Question:** Is county-level air quality (median AQI) associated with adult asthma prevalence across U.S. counties, and what does the pattern suggest for the Los Angeles area?
- **Answer:** No. Across 769 counties, air quality shows no meaningful relationship with adult asthma. County size and state matter more, so county AQI alone is a poor guide for targeting asthma programs.

Pyramid Principle

Governing thought: Air quality does not explain adult asthma across U.S. counties, so decisions need finer-grained data.

1. **Counties with worse air do not report more asthma.**
    - Support: correlation of -0.04, 10.67% vs. 10.61% asthma in the lowest and highest AQI quartiles, and every statistical test agrees.
2. **County size and state explain more than air quality does.**
    - Support: larger counties report less asthma (r = -0.26), and adding state raises explained variation from 9% to 67%.
3. **County averages can mislead, as Los Angeles shows.**
    - Support: high AQI (62) but below-average asthma (9.1%), and the next step is neighborhood-level data.

Storyline (Horizontal Flow)

Read top to bottom, these action titles should tell the full story on their own.

1. Title: Does Air Quality Explain Adult Asthma? A County-Level Analysis of CDC and EPA Data
2. We tested whether U.S. counties with cleaner air report less adult asthma
3. Air quality does not explain adult asthma across counties; size and state matter more
4. We compared 769 counties by joining CDC asthma estimates with EPA air quality data
5. Counties with worse air do not report more asthma (correlation of -0.04)
6. Every test agrees: asthma is the same in the lowest- and highest-AQI counties
7. County size and state matter more than air quality
8. Los Angeles has high AQI but below-average asthma, so county averages can mislead
9. These results show the limits of county data, not that air pollution is harmless
10. Neighborhood-level data, income measures, and child asthma are the next steps

Appendix A1: Most counties sit near 10.6% asthma and a median AQI of 41
Appendix A2: Name matching and suppressed estimates reduced 827 monitored counties to 769 rows

Slide-by-Slide Outline (Vertical Flow)

Each slide below lists its dot (action title) and its dashes (body). Keep every slide to three or four dashes and one visual. Put "Source: CDC PLACES (2025), EPA AirData (2026), author analysis" in the footer of each data slide.

Slide 1: Title

- **Does Air Quality Explain Adult Asthma? A County-Level Analysis of CDC and EPA Data**
    - Presenter, TruBridge Health Analytics Externship, and date

Slide 2: Context (Situation, Complication, Question)

- **We tested whether U.S. counties with cleaner air report less adult asthma**
    - Situation: asthma affects millions of adults, and air pollution is a known trigger
    - Complication: it is unclear whether county air quality tracks asthma, and the Los Angeles area has long faced air-quality challenges
    - Question: is county-level AQI associated with adult asthma prevalence?

Visual: three stacked text boxes (Situation, Complication, Question). No chart.

Slide 3: Answer First (Executive Summary)

- **Air quality does not explain adult asthma across counties; size and state matter more**
    - Counties with worse air do not report more asthma (r = -0.04)
    - County size and state differ more than air quality does
    - Los Angeles shows why county averages can mislead
    - Recommendation: add neighborhood-level, income, and child data before targeting programs

Visual: three numbered takeaway callouts and a one-line recommendation.

Slide 4: Data and Approach

- **We compared 769 counties by joining CDC asthma estimates with EPA air quality data**
    - CDC PLACES (2025): modeled adult asthma prevalence for 3,143 counties
    - EPA Annual AQI (2026): median AQI for 827 monitored counties
    - Merged on state and county, leaving 769 rows with both values
    - Methods: correlation, group comparisons, and regression with state effects

Visual: a simple flow of two data sources, then merge, then 769 counties, then analysis. Cleaning details are in Appendix A2.

Slide 5: Air Quality vs. Asthma

- **Counties with worse air do not report more asthma (correlation of -0.04)**
    - The trend line is flat across median AQI values from 3 to 72
    - Unhealthy-day share (-0.11), PM2.5 days (0.00), and ozone days (-0.03) show the same
    - The highest asthma in the sample (Apache County, Arizona, 14.3%) has one of the lowest AQI values (8)

Visual: Figure 2, the scatter plot of asthma vs. median AQI, with Los Angeles County highlighted.

Slide 6: Statistical Tests

- **Every test agrees: asthma is the same in the lowest- and highest-AQI counties**
    - Lowest vs. highest AQI quartile: 10.67% vs. 10.61% (t-test $p = 0.48$, Mann-Whitney $p = 0.82$)
    - By dominant pollutant (ANOVA $p = 0.35$) and by above-median asthma (chi-square $p = 0.37$): no difference
    - Regression: AQI is not significant ($p = 0.19$), even within states ($p = 0.48$)

Visual: Figure 3, the box plots by AQI quartile. Add a small version of Table 1 if space allows.

Slide 7: County Size and State

- **County size and state matter more than air quality**
    - Larger counties report slightly less asthma ($r = -0.26$)
    - Adding state raises explained variation from 9% to 67%, while AQI stays non-significant
    - The highest county averages are in Maine, Oregon, West Virginia, Massachusetts, and Oklahoma

Visual: Figure 4, the scatter plot of asthma vs. county population (log scale).

Slide 8: The Los Angeles Area

- **Los Angeles has high AQI but below-average asthma, so county averages can mislead**
    - Los Angeles County: AQI 62 and asthma 9.1%, compared with a 10.6% sample average. Orange (9.0%) and San Diego (8.9%) look similar
    - Northern counties such as Trinity and Humboldt have the highest asthma (11.0%) at moderate AQI
    - Los Angeles research shows asthma clusters by neighborhood, which a county average hides

Visual: Figure 5, the scatter plot of 40 California counties with Los Angeles highlighted.

Slide 9: Limitations

- **These results show the limits of county data, not that air pollution is harmless**
    - County averages hide neighborhood exposure, and PLACES covers adults only

    - Only monitored counties (827) are included, which skews toward larger counties
    - Income, race, and smoking are missing, and AQI is not long-term exposure

Visual: none, or three simple icons.

Slide 10: Recommendations and Next Steps

- **Neighborhood-level data, income measures, and child asthma are the next steps**
    - Repeat the analysis with census-tract data in Los Angeles County
    - Add income, poverty, and race and ethnicity from the American Community Survey
    - Study childhood asthma and use multi-year PM2.5 data
    - Avoid targeting asthma programs by county AQI alone

Visual: a four-step roadmap graphic.

Appendix A1: Distributions

- **Most counties sit near 10.6% asthma and a median AQI of 41**
    - Asthma averages 10.6% with a range of 8.0% to 14.3%
    - Median AQI centers at 41, with a tail of cleaner counties
    - Population is heavily right-skewed (median about 120,500), so the analysis uses its log

Visual: Figure 1, the histograms of asthma and median AQI.

Appendix A2: Data Cleaning and Merging

- **Name matching and suppressed estimates reduced 827 monitored counties to 769 rows**
    - 27 AQI records matched no PLACES county (naming differences, Alaska boroughs, Puerto Rico, and one Mexico record)
    - 34 matched counties have no asthma estimate in PLACES
    - Three counties (Baltimore, Fairfax, and Roanoke) matched both a city and a county, so they appear twice; removing them does not change the results
    - Derived variables: unhealthy-day share, dominant pollutant, AQI quartiles, and log population

Visual: none.